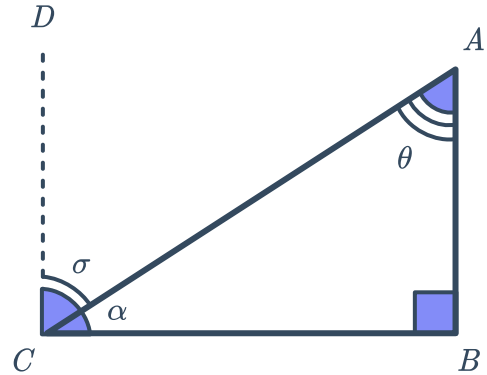


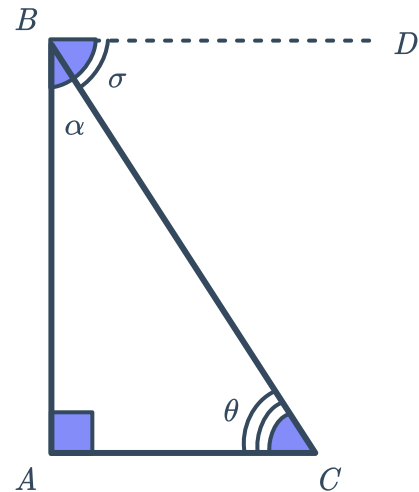
Angles of elevation and depression



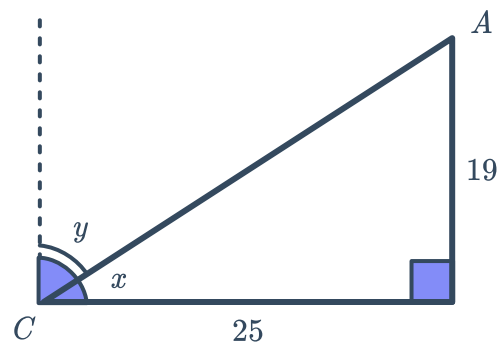
- 1 A man standing at point C , is looking at the top of a tree at point A . Identify the angle of elevation in the following figure:



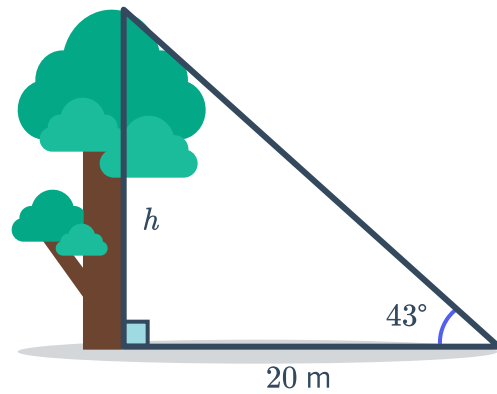
- 2 A man standing at the top of the tower at point B , is looking at the ground at point C . Identify the angle of depression in the following figure:



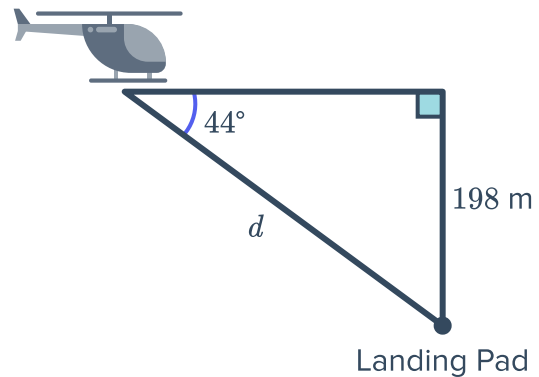
- 3 Find x , the angle of elevation to point A from point C . Round your answer to two decimal places.



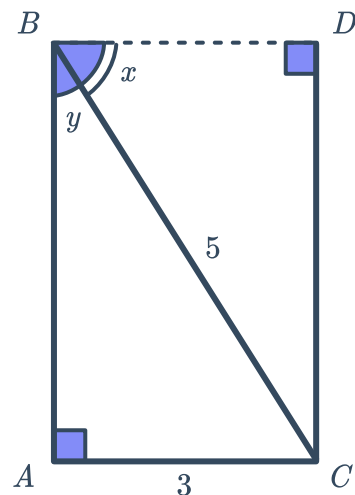
- 4 Sally measures the angle of elevation to the top of a tree from a point 20 m away to be 43° .
Find the height of the tree, h , to the nearest whole number.



- 5 A helicopter is flying at an altitude of 198 m. Its landing pad is at an angle of depression of 44° .
Determine the distance, d , between the helicopter and the landing pad. Round your answer to the nearest whole number.

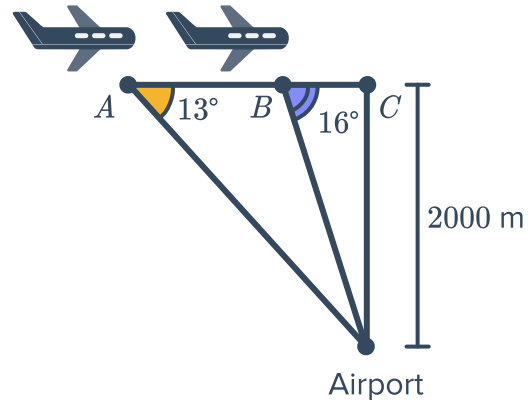


- 6 Find x , the angle of depression from point B to point C in the given diagram:
Round your answer to two decimal places.



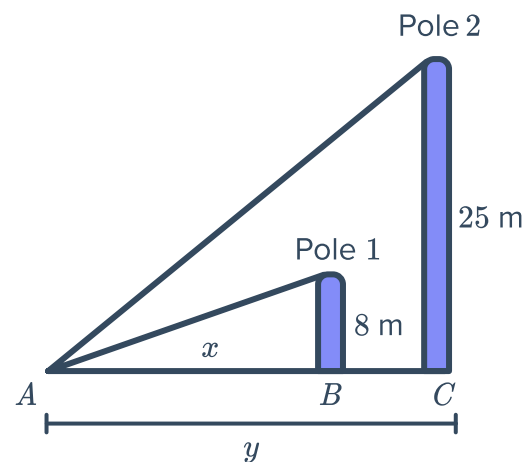
- 7 A fighter jet, flying at an altitude of 2000 m is approaching an airport. The pilot measures the angle of depression to the airport to be 13° . One minute later, the pilot measures the angle of depression again and finds it to be 16° .

- Find the distance AC , to the nearest metre.
- Find the distance BC , to the nearest metre.
- Hence, find the distance covered by the jet in that one minute, to the nearest metre.

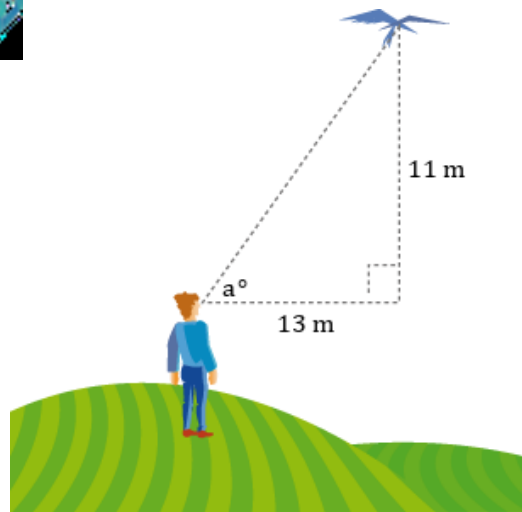


- 8 A man stands at point A looking at the top of two poles. Pole 1 has a height 8 m and an angle of elevation of 34° from point A . Pole 2 has a height 25 m and an angle of elevation of 57° from point A .

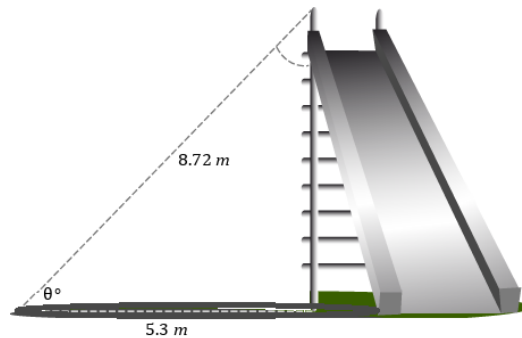
- Find x , the distance from A to B , to two decimal places.
- Find y , the distance from A to C , to two decimal places.
- Hence, find BC , the distance between the two poles in meters. Round your answer to one decimal place.



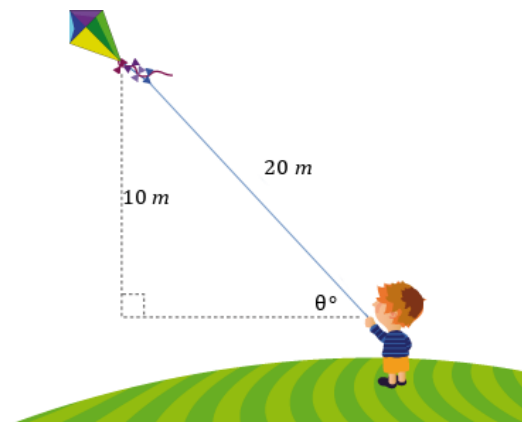
- 9 The person in the picture sights a paraglider above him. Find a to two decimal places.



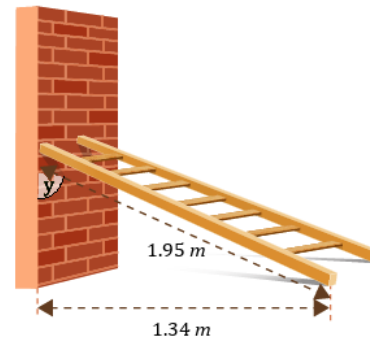
- 10 A slide casts a shadow 5.3 m along the ground. The distance between the tip of the shadow and the top of the slide is 8.72 m. Find θ to two decimal places.



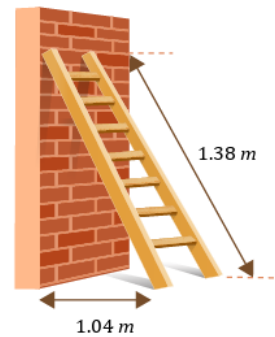
- 11 A boy flying his kite releases the entire length of his string which measures 20 m, so that the kite is 10 m above him. If the angle the string makes with the horizontal ground is θ , find θ to two decimal places.



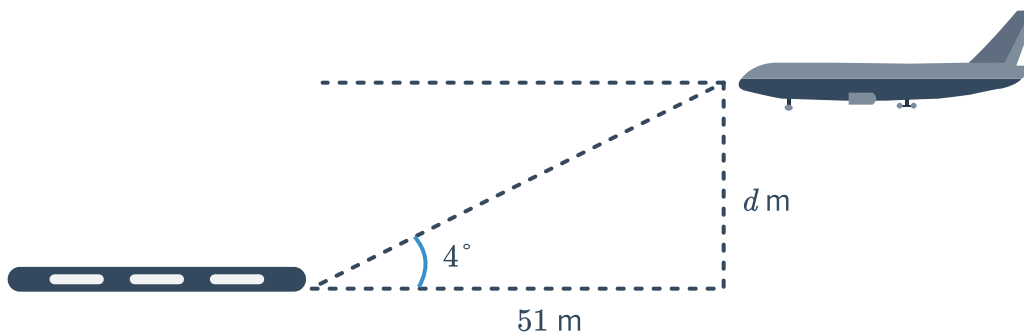
- 12 A ladder measuring 1.95 m in length is leaning against a wall. If the angle the ladder makes with the wall is y° , find y to two decimal places.



- 13 A ladder measuring 1.38 m in length is leaning against a wall. If the angle the ladder makes with the ground is y° , find y to two decimal places.

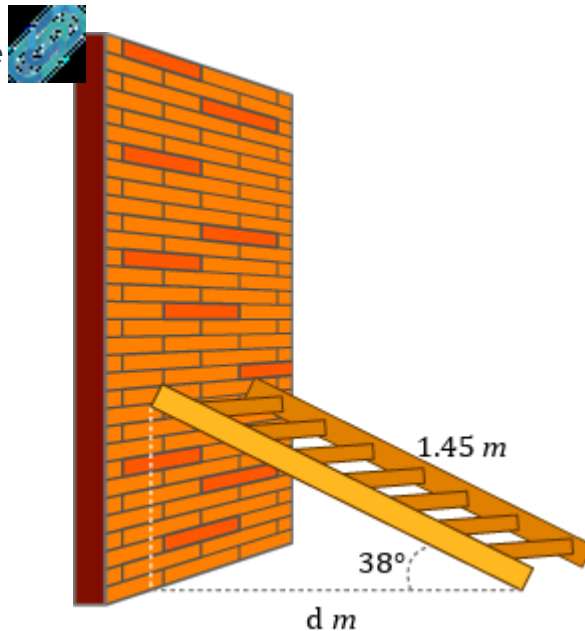


- 14 The final approach of an airplane when landing requires an angle of descent of about 4° . If the plane is directly above a point 51 m from the start of the runway, find d , the height of the plane above the ground to the nearest meter.

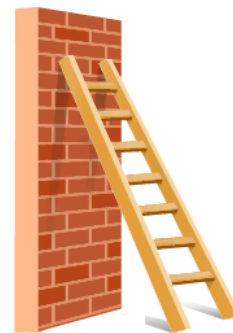


The final approach of an airplane when landing requires an angle of descent of about 4° .

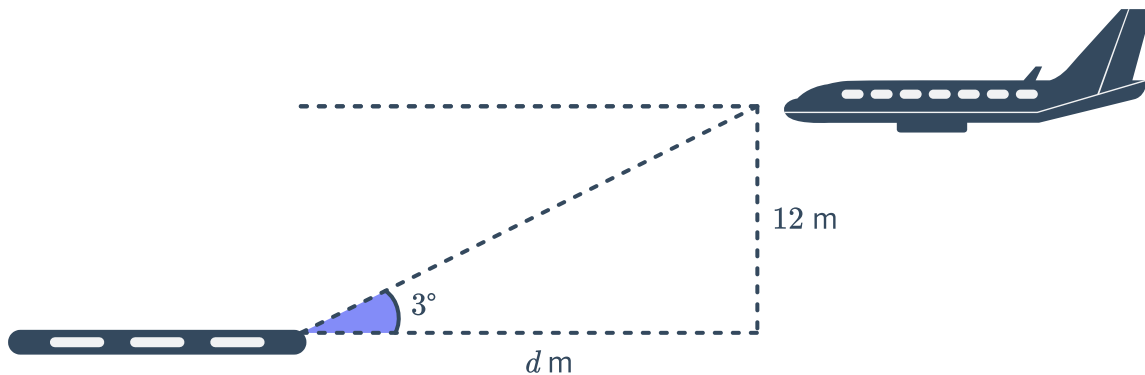
- 15 If d is the distance between the base of the wall and the base of the ladder, find the value of d to two decimal places.



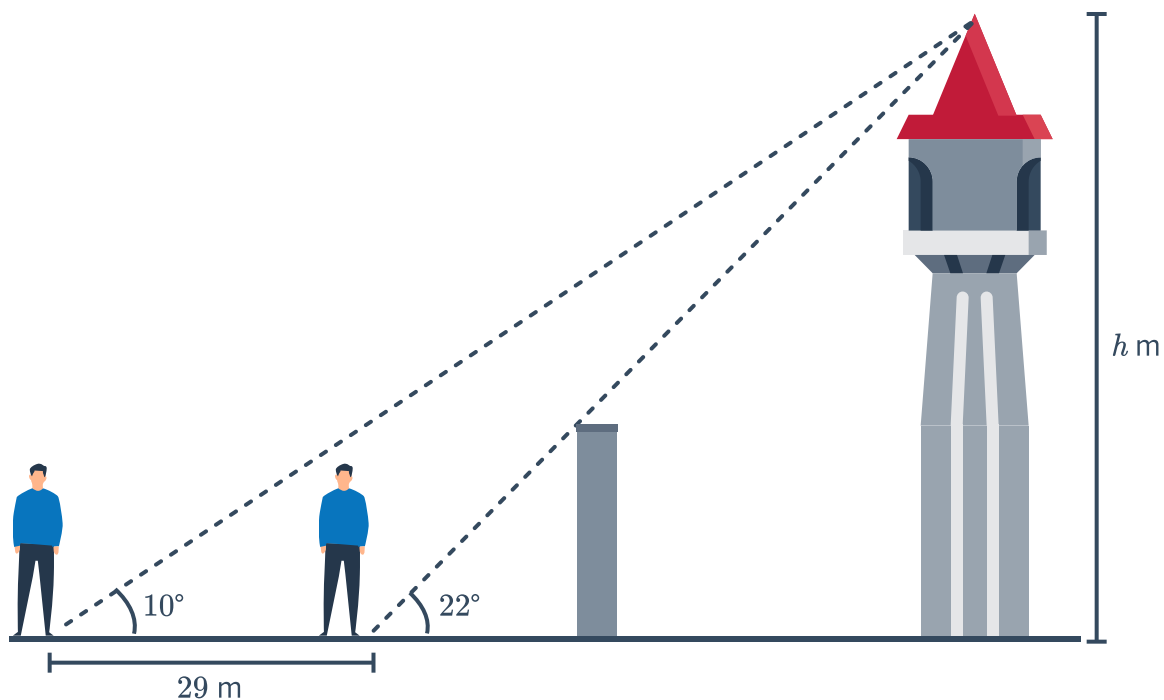
- 16 A ladder of length 157 cm needs to make an angle with the wall of between 10° and 20° for it to be safe to stand on.
- a If the base of the ladder is 85 cm from the wall, and the angle the ladder makes with the wall is x° , find x to two decimal places.
 - b If the base of the ladder is 17 cm from the wall, and the angle the ladder makes with the wall is y° , find y to two decimal places.
 - c If the base of the ladder is 28 cm from the wall, and the angle the ladder makes with the wall is z° , find z to two decimal places.
 - d At which length from the wall is the ladder safe?



- 17 The final approach of an airplane when landing requires the pilot to adjust the angle of descent to about 3° as shown in the diagram below. If the plane is 12 m above the runway and has d m until touchdown, find d to the nearest meter.



- 18 A safety fence is constructed to protect tourists from the danger of an eroding castle toppling down. The surveyor takes an angle measurement to the top of the tower of 10° . She then walks 29 m towards the tower and takes another reading of 22° . Find the value of h to the nearest meter.



- 19 Amelia measures the angle of elevation to the top of a tree from a point, 29 m away from the base, to be 31° . Find the height of the tree, h , to the nearest meter.



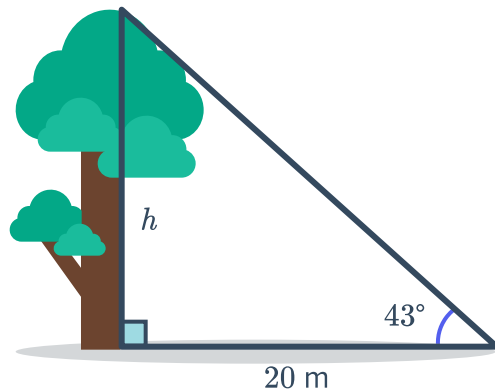
- 20** The angle of elevation from an observer to the top of a tree is 18° . If the distance between the tree and the observer is d m and the tree is known to be 3.53 m high, find the value of d to two decimal places.
- 21** At a certain time of the day a light post, 6 m tall, has a shadow of 5.8 m. If the angle of elevation of the sun at that time is θ° , find θ to two decimal places.
- 22** From the top of a rocky ledge 188 m high, the angle of depression to a boat is 13° . If the boat is d m from the foot of the cliff, find the value of d correct to two decimal places.
- 23** Lisa is on a ship and observes a lighthouse on a cliff in the distance. The base of the cliff is 906 m away from the ship, and the angle of elevation of the top of the lighthouse from Lisa is 16° .
- a** If the top of the lighthouse is x m above sea level, find the value of x correct to two decimal places.
 - b** If the lighthouse is 21 m tall, how tall is the cliff? Round your answer to two decimal places.
- 24** Buzz is standing 49 m from a building and measures the angle of elevation of the top of the building to be 23° .
- a** If the difference in height between the top of the building and Buzz's eye is h m, find the value of h correct to two decimal places.
 - b** If Buzz's eye is 135 cm from the ground, what is the height of the building? Round your answer to one decimal place.
- 25** From the top of a cliff, 47 m above sea level, two boats are observed to be in the same direction. Their angles of depression from the top of the cliff are 12° and 23° .
- a** Find the distance between the base of the cliff and the boat furthest away, correct to two decimal places.
 - b** Find the distance between the base of the cliff and the closest boat, correct to two decimal places.
 - c** Hence, find the distance between the two boats in meters, correct to one decimal place.
- 26** A ship dropped anchor off the coast of a resort. The anchor fell 65 m to the sea bed. During the next 5 hours, the ship drifted 120 m. Calculate the angle of depression, x , between the anchor line and the surface of the water, rounded to the nearest degree.



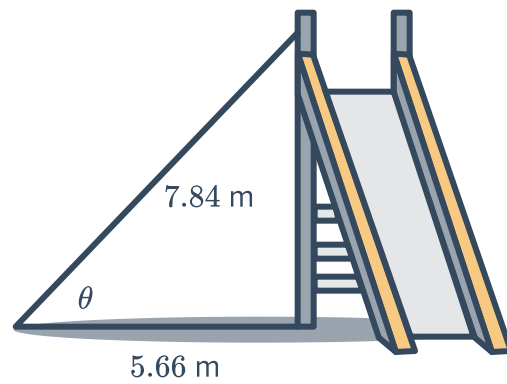
- 27** A 13.7 m long string of lights joins the top of a tree to a point on the ground. If the tree is 3.7 m tall, find θ , the angle the string of lights would make with the tree, rounded to two decimal places.
- 28** Jack is standing at the tip of a tree's shadow and knows that the angle from the ground to the top of the tree is 34° . If Jack is standing 29 m away from the base of the tree, find the height of the tree to two decimal places.
- 29** The angle of elevation to the top of a 22 m high statue is 54° from a point, A , due west of the statue. The point B is located 60 m due south of point A .
- Find the distance, x , from point A to the base of the statue, correct to two decimal places.
 - Hence, find y , the distance from point B to the base of the statue, correct to one decimal place.
 - Hence, find θ , the angle of elevation from point B to the top of the statue, correct to the nearest degree.

Additional questions

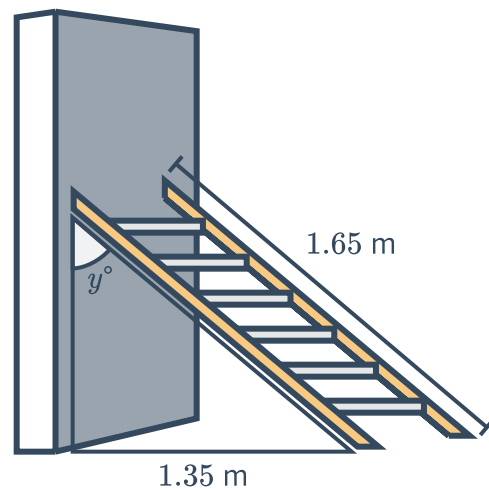
- 30** Aicha measures the angle of elevation to the top of a tree from a point 20 m away to be 43° . Find the height of the tree, h , to the nearest meter.



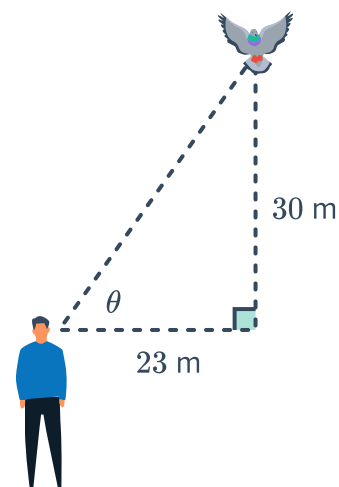
- 31 A slide casts a shadow 5.66 m along the ground. The distance between the tip of the shadow and the top of the slide is 7.84 m. Find θ , rounding to two decimal places.



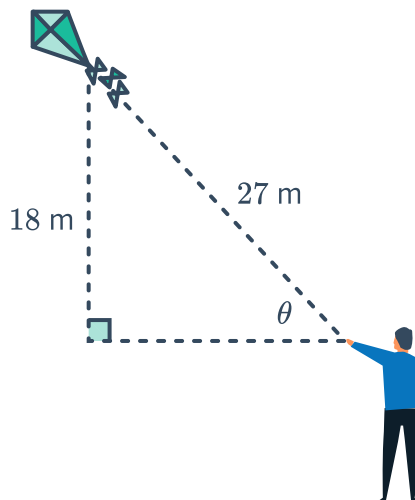
- 32 A ladder measuring 1.65 m in length is leaning against a wall. If the angle the ladder makes with the wall is y° , find y , rounding to two decimal places.



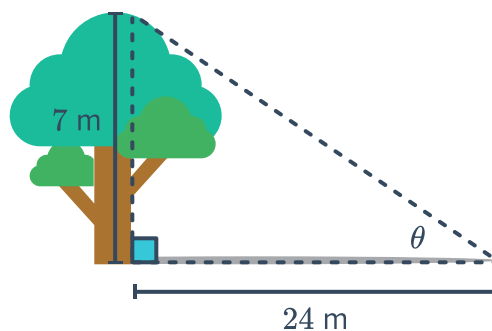
- 33 Xander sights a pigeon above him. Find θ , rounding to two decimal places.



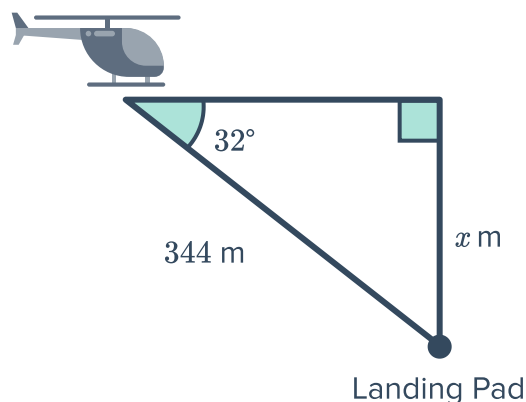
- 34** Saul is flying his kite and releases the entire length of his string which measures 27 m, so that the kite is 18 m above him. If the angle the string makes with the horizontal ground is θ , find θ , rounding to two decimal places.



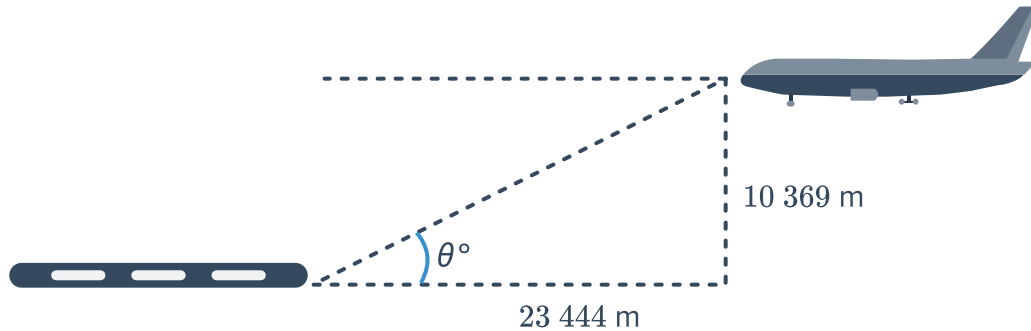
- 35** During a particular time of the day, a tree casts a shadow of length 24 m. The height of the tree is estimated to be 7 m. Find the angle θ , formed by the length of the shadow and the arm extending from the edge of the shadow to the top of the tree. Round your answer to two decimal places.



- 36** A helicopter is 344 m away from its landing pad. If the angle of depression to the landing pad is 32° , find x , the height of the helicopter above the ground, to the nearest meter.



- 37 The airtraffic controller is communicating with a plane in flight approaching an airport for landing. The plane is 10 369 m above the ground and is still 23 444 m from the runway. If θ° is the angle at which the plane should approach, find θ , rounding to the nearest tenth.



- 38 Glenda is riding her pushbike up a hill that has an incline of 5° . If she rides her bike 817 ft up the hill, find d , the horizontal distance from where Glenda started, rounding to the nearest foot.
- 39 A ship dropped anchor off the coast of a resort. The anchor fell 83 ft to the sea bed. During the next 4 hours, the ship drifted 105 ft. Find x , the angle between the anchor line and the surface of the water, rounding to the nearest degree.
- 40 A 27.3 m long cable joins the top of a flagpole to a point on the ground. If the flagpole is 7.4 m tall, find θ , the angle the cable would make with the flagpole, rounding to two decimal places.
- 41 From the cockpit of an airplane flying at an altitude of 3000 m, the angle of depression to the airport is 57° . The airplane continues to fly in the same straight line, and after a few minutes the angle of depression to the airport becomes 67° .
- Sketch a diagram of the situation.
 - The horizontal distance between the cockpit at the first sighting and the point directly above the airport is x m. Find the value of x , rounding your answer to the nearest tenth.
 - The horizontal distance between the cockpit at the second sighting and the point directly above the airport is y m. Find the value of y , rounding your answer to the nearest tenth.

42 Valentina and Mateo are on opposite sides of a valley. They both look down into the valley between them and see a large black bear. Valentina is 75 ft above the valley and sees the bear at an angle of depression of 49° . Mateo is 59 ft higher than Valentina and the angle of elevation from the bear to Mateo is 40° .



- a Assuming Valentina and Mateo are both standing at the edges of cliffs at each side of the valley, determine the width of the valley to the nearest foot.
- b Calculate the distance between Valentina and Mateo. Round your answer to the nearest tenth